



UNIVERSITY of
RWANDA

COLLEGE OF SCIENCE
AND TECHNOLOGY

DEPARTMENT OF MATHEMATICS

FINAL EXAMINATION - ACADEMIC YEAR 2023/2024

First YEAR (CE, TRE, WREE, SGE)-THIRD TRIMESTER

MAT1263: Advanced Calculus for Engineers

Date: 13/05/2024

Time: 2 hours

INSTRUCTIONS

1. This paper has SECTION A and SECTION B.
2. Questions in SECTION A are compulsory
3. Answer any TWO questions from SECTION B.
4. No written materials allowed in examination room.
5. Write all your answers in the answer booklet provided.
6. Do not forget to write your registration number.

Section A

Question 1 (20marks)

1. When a cake is removed from an oven, its temperature is measured at 300°F . Three minutes later its temperature is 200°F . How long will it take for the cake to cool off to a room temperature of 70°F ? (2marks)
2. Given that $y_1(x) = x$ is a solution of the homogeneous linear differential equation

$$x^2 y'' - xy' + y = 0, \quad 0 < x < \infty$$

- (a) Use reduction of order to obtain a second solution, $y_2(x)$. Does the pair $\{y_1, y_2\}$ form a fundamental set of solutions of this differential equation? (3marks)
- (b) If $\{y_1, y_2\}$ form a fundamental set of solutions, solve the initial-value problem (2marks)

$$x^2 y'' - xy' + y = 0, \quad y(1) = 3, \quad y'(1) = 8.$$

3. If $y = f(x)$ is the solution of $\frac{d^2 y}{dx^2} = 0$ with the boundary conditions $y(0) = 5$, $y'(0) = 2$; find $f(15)$ (3marks)
4. (a) State the conditions under which Laplace Transform exists for a function $f(t)$ (1mark)
(b) What is meant by piecewise continuous (1mark)

5. Evaluate $\mathcal{L}^{-1} \left\{ \frac{-2s+6}{s^2+4} \right\}$ (2marks)

6. Evaluate the line integral from the origin to the point $P(1,1,1)$.

$$\int (2xyz \vec{i} + x^2 z \vec{j} + x^2 y \vec{k}) d\vec{r}$$

(2marks)

7. Suppose that

$$\nabla \phi(x, y) = (y^2 + 2x + 4)\vec{i} + (2xy + 4y - 5)\vec{j}$$

Find the function $\phi(x, y)$ (2marks)

8. Find the rate of change of $f(x, y) = x^3 + 3y^2$ at the point $(1, 2)$ in the direction of $\vec{u} = 3\vec{i} + 4\vec{j}$ (2marks)

Section B

Question 2

1. Solve the non homogeneous initial value problem (7marks)

$$\frac{d^2 y}{dt^2} + y = t, \quad y(0) = y'(0) = 1$$

2. Use the Laplace transform and the inverse Laplace transform to solve the following initial-value problem (8marks)

$$\begin{cases} x' = x + 3y \\ y' = 3x + y \end{cases} \quad x(0) = 5, \quad y(0) = 1.$$

Question 3

1. Verify the Green formula for $\oint_C (2xy - x^2) dx + (x + y^2) dy$, where C is the curve of the region of the xy plane bounded by $y = x^2$ and $x = y^2$. (9marks)
2. Evaluate $\mathcal{L} \left\{ \int_0^t e^{-u} \cos u \, du \right\}$ (6marks)

Question 4

1. Sketch the region of integration, reverse the order of integration, and evaluate the integral (10marks)

$$\int_0^2 \int_x^2 2y^2 \sin(xy) \, dy \, dx$$

2. Show that $\vec{F} = (6xy + z^3)\vec{i} + (3x^2 - z)\vec{j} + (3xz^2 - y)\vec{k}$ is irrotational (5marks)